Instruction

Mock Project Overview: *Falcon* University Admission Pre-Assessment

+ Objective

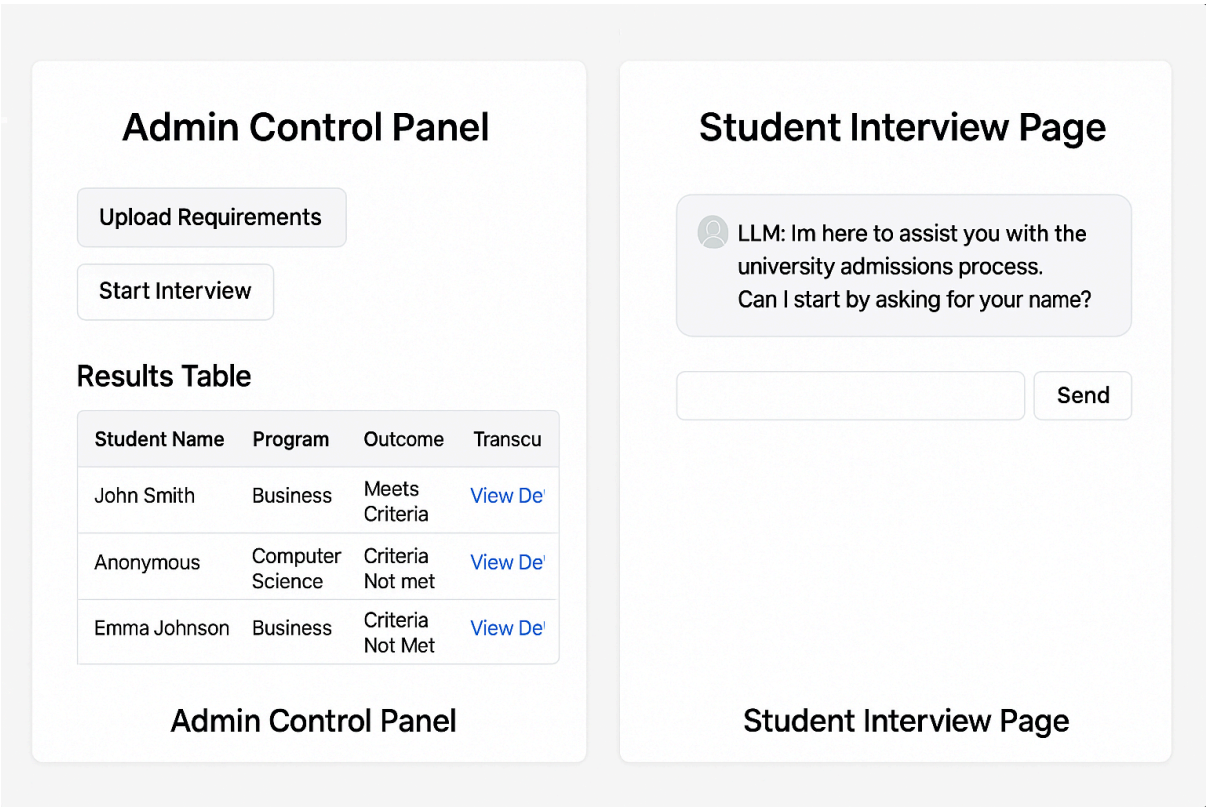
A- Build a lightweight web app that simulates an admission officer, powered by an LLM assistant, to guide applicants through a chat-based eligibility check. The bot would:

(a) chats with a student,
(b) retrieves knowledge from an uploaded PDF. (Use this)
(c) performs the eligibility check based on preset rules, and
(d) shows the outcome: Eligible to Proceed

B- After finishing step A, [lease think about the product [ ] and answer the product thinking questions..

⏱️ **Deadline:** Wednesday, **04/22/2025**

+ Workflow



Step 1: Build Admin page

1. **Login Page with Authentication**
2. **Dashboard Page** → Includes
    1. *"Upload Requirements" Button*
    2. Result Table (check next section)
    3. *"Start interview" Button*

3. **Admin uploads <u>a PDF file</u> that includes the chatbot's questions and requirements** → Use knowledge base as exmple→ Once uploaded, the backend parses text and prepares it as the knowledge base for chat.

4. Create the applicants database

   Column Description:

   - Student Name Name entered during interview (or "Anonymous" if not provided).
   - Program Program applied for (Business or Computer Science)
   - Outcome Final eligibility result: Meets Criteria / Criteria Not Met

- Rule Summary Short text explanation
- Transcript Link Button or link to open the View Details drawer with chat transcript.
- Transcription Conversation between LLM and user

Step 2: Build User page

1. **"Start Interview"** Button→ Clicking the button opens the chat panel.
2. **LLM Q&A and Eligibility Check** →
    - LLM greets the student and begins asking eligibility questions.
    - Student replies in text.
    - Once data is collected, the LLM evaluates the rules, tells the student the result, and closes the chat.

Step 3: Update the admin page with the user's input

1. **Table Updates** → The results table updates with the gathered information

Step 4: Deployment

Deployment options:

i. *Local*
ii. *Bonus:* *Deploy webapp on AWS or other cloud providers free accounts (Vercel,Railway,etc)

+ Suggested Stack

- **Frontend**: Next.js or Vite + Tailwind + Vercel

- **Backend**: Node.js , FastAPI (Python)
- **LLM**: OpenAI and Lang Chain (RAG + tool/function-calling)
- **Database**: PostgreSQL, or AWS stacks (Bonus)
- **Dev/Deploy**: Docker Compose for local; AWS for optional cloud deployment(Bonus)

+ Evaluation Criteria

- Functionality (chat + eligibility check working end-to-end).
- Code quality (clean structure, modular backend).
- LLM/RAG use (retrieval applied meaningfully).
- UI/UX (simple, clean dashboard; transcript and reasons visible).

+ Submission Requirements

- Share a GitHub repo with https://github.com/rmyzm
- Short Loom demo (2–3 minutes: demo and explanation
    - **Showcase the demo**
    - **Mention the tech stack you used**
    - **Explain where and how the LLM or Copilots were used to build the webpage**
- README with setup instructions.
- README includes responses to product thinking questions.

Knowledge base

📚 Knowledge Base: University Application

🎓 Business Program – Requirements

1. **Age Requirement**: Applicants must be at least 16 years old at the time of application.

2. **High School Completion**: A high school diploma or equivalent certification is required.

3. **Minimum GPA**: A GPA of **3.0 on a 4.0 scale** or higher is required.

4. **Standardized Tests**: Applicants should have taken either the SAT or ACT.

5. **SAT Score Threshold**: A minimum SAT score of **1200** is expected.

6. **ACT Score Threshold**: If applying with ACT, a score of at least **27** is required.

7. **English Proficiency (International Students)**: Non-native English speakers must provide TOEFL (≥95), IELTS (≥7.0), or Duolingo (≥120).

8. **Personal Statement**: A written statement of purpose must be submitted.

9. **Recommendation Letter**: At least one teacher or counselor recommendation is mandatory.

10. **Application Completeness**: All required documents (transcripts, test scores, essay, recommendation) must be included at the time of submission.

💻 Computer Science Program – Requirements

1. **Age Requirement**: Applicants must be at least 16 years old at the time of application.

2. **High School Completion**: A high school diploma or equivalent certification is required.

3. **Minimum GPA**: A GPA of **3.3 on a 4.0 scale** or higher is required.

4. **Mathematics Proficiency**: Applicants should have completed advanced math courses (e.g., calculus, algebra).

5. **Standardized Tests**: Applicants should have taken either the SAT or ACT.

6. **SAT Score Threshold**: A minimum SAT score of **1300** is expected.

7. **ACT Score Threshold**: If applying with ACT, a score of at least **29** is required.

8. **English Proficiency (International Students)**: Non-native English speakers must provide TOEFL (≥95), IELTS (≥7.0), or Duolingo (≥120).

9. **Personal Statement**: Applicants should explain their interest in computing and problem-solving.

10. **Recommendation Letter**: At least one recommendation from a STEM teacher or mentor is required.

Product Thinking

★ Product thinking assessment

1. Stakeholder Alignment

Who are the key stakeholders for this admission pre-assessment tool, and how would you ensure their needs are reflected in the product roadmap?

2. Success Metrics

List 2–3 metrics you would use to measure the success of the product.

3. Risk Awareness

What is one major risk of using an AI assistant for eligibility screening, and how would you mitigate it?